

Richard Li

Email: Lichard@cs.washington.edu

Website: <https://Lichard49.github.io>

Employment

- 2025 - Present **University of Washington** - *Postdoctoral Fellow*
Mentors: James Fogarty (DUB Group), Shwetak Patel (UbiComp Lab), and Cindy Lin (School of Medicine)
- 2019 - 2025 **University of Washington** - *Graduate Research Assistant*
Mentors: James Fogarty (DUB Group) and Shwetak Patel (UbiComp Lab)
- 2022 **Microsoft Research** - *Research Intern*
Mentors: Teddy Seyed, Nicolai Marquardt, Nathalie Riche, and Ken Hinckley (EPIC Team)
- 2021 **Microsoft Research** - *Research Intern*
Mentors: Eyal Ofek, Nicolai Marquardt, and Ken Hinckley (EPIC Team)
- 2020 **Microsoft Research** - *Research Intern*
Mentors: Alex Fiannaca and Ed Cutrell (Ability Team)
- 2019 - 2020 **X, the Moonshot Factory (Google [x])** - *Research Scholar*
Mentors: JB Berent and Yash Ahluwalia
- 2018 - 2019 **Google Research** - *Research Scholar*
Mentors: Dimitri Kanvesky, Thad Starner, Dick Lyon
- 2017 - 2019 **Georgia Institute of Technology** - *Graduate Research Assistant*
Mentors: Thad Starner and Gregory Abowd
- 2017 **Nvidia Research** - *Research Intern*
Mentor: Kaan Akşit
- 2015 **Amazon.com** - *Software Engineering Intern*
- 2014 **Intel** - *Research and Development Intern*
- 2013 **Broadcom** - *Software Configuration Management Intern*

Education

- 2019 - 2025 **University of Washington**
Ph.D. in Computer Science & Engineering
Advisors: James Fogarty and Shwetak Patel
- 2016 - 2018 **Georgia Institute of Technology**
M.S. in Computer Science
Advisors: Gregory Abowd and Thad Starner
- 2012 - 2016 **Georgia Institute of Technology**
B.S. in Computer Science, minor in Music Performance (clarinet)

Awards

- 2025 **Young Researcher** - *Heidelberg Laureate Forum*
- 2025 **CHI 2025 Best Paper Award** - *CHI Conference on Human Factors*
- 2024 **Travel Grant to UbiComp/ISWC 2024** - *UbiComp/ISWC*
- 2024 **I-Corps Participant** - *UW NSF Innovation Corps Site*

- 2024 **Innovation Gap Fund (~\$10k)** - UW CoMotion Labs
- 2023 **Inventor of the Year Finalist** - UW School of Medicine
- 2023 **Husky 100** - University of Washington
- 2023 **Population Health Pilot Grant (~\$100k)** - UW Population Health Initiative
- 2019 **Futuremaker Fellowship (~\$50k)** - Siemens
- 2017 **Runner-Up Master's Thesis** - GT Three Minute Thesis Competition
- 2017 **Outstanding Master's Thesis** - GT College of Computing
- 2017 **Best Computing Poster Award** - GT Career, Research, and Innovation Development Conference
- 2015 **Travel Grant to CSCW 2015** - GT GVU Center
- 2015 **President's Undergraduate Research Award** - GT Undergraduate Research Opportunities Program
- 2014 - 2016 **HOPE Scholarship** - GT Office of Scholarships and Financial Aid

Refereed Publications

- 2025 **Beacon: At-Home, Self-Administration of a Test for Critical Flicker Frequency as a Screening Test for Minimal Hepatic Encephalopathy**
AASLD The Liver Meeting
 Philip Vutien*, Richard Li*, Ravi Karkar, James Fogarty, Sean A Munson, Sabrina Omer, Michael Yacoub, James Kashima, George N Ioannou
 **Oral Presentation**
- 2025 **What Motivates Bladder Cancer Patients to be Active? A Qualitative Study Assessing Attitudes Towards Physical Activity and Digital Health Technologies**
Urologic Oncology: Seminars and Original Investigations
 Alex Zhu, Jessica Sher, Richard Li, Otari Ioseliani, Leah Cantor, Elena G Brewer, Karla Landis, David Bridges, Sean A Munson, Hanna Hunter, Cindy Lin, Sarah P Psutka
- 2025 **Deploying and Examining Beacon for At-Home Patient Self-Monitoring with Critical Flicker Frequency**
ACM CHI Conference on Human Factors in Computing Systems
Richard Li, Philip Vutien, Sabrina Omer, Michael Yacoub, George Ioannou, Ravi Karkar, Sean A Munson, James Fogarty
 **Best Paper Award**
- 2025 **ECG Necklace: Low-power Wireless Necklace for Continuous ECG monitoring**
ACM CHI Conference on Human Factors in Computing Systems
 Qiuyue Xue, Eric Steven Martin, Jiaqing Liu, Ruiqing Wang, Antonio Glenn, Richard Li, Vikram Iyer, Shwetak Patel
- 2023 **Evaluating a Novel, Portable, Self-Administrable Device ("Beacon") that Measures Critical Flicker Frequency as a Test for Hepatic Encephalopathy**
ACG American Journal of Gastroenterology
 Philip Vutien*, Richard Li*, Ravi Karkar, Sean A Munson, James Fogarty, Kara Walter, Michael Yacoub, George N Ioannou
- 2023 **AdHocProx: Sensing Mobile, Ad-Hoc Collaborative Device Formations using Dual Ultra-Wideband Radios**
ACM CHI Conference on Human Factors in Computing Systems
Richard Li, Teddy Seyed, Nicolai Marquardt, Eyal Ofek, Steve Hodges, Mike Sinclair, Hugo Romat, Michel Pahud, Jatin Sharma, William AS Buxton, Ken Hinckley, Nathalie Riche
- 2022 **Food, Mood, Context: Examining College Students' Eating Context and Mental Well-Being**

- ACM Transactions on Computing for Healthcare*
 Mehrab Bin Morshed, Samruddhi Shreeram Kulkarni, Koustuv Saha, [Richard Li](#), Leah G Roper, Lama Nachman, Hong Lu, Lucia Mirabella, Sanjeev Srivastava, Kaya De Barbaro, Munmun De Choudhury, Thomas Plötz, Gregory Abowd
- 2022 **SilentSpeller: Towards Mobile, Hands-Free, Silent Speech Text Entry using Electropalatography**
ACM CHI Conference on Human Factors in Computing Systems
 Naoki Kimura, Tan Gemicioglu, Jonathan Womack, [Richard Li](#), Yuhui Zhao, Abdelkareem Bedri, Zixiong Su, Alex Olwal, Jun Rekimoto, Thad Starner
- 2021 **Understanding the Design Space of Mouth Microgestures**
ACM Designing Interactive Systems Conference
 Victor Chen, Xuhai Xu, [Richard Li](#), Yuanchun Shi, Shwetak Patel, Yuntao Wang
- 2021 **Augmented Silkscreen: Designing AR Interactions for Debugging Printed Circuit Boards**
ACM Designing Interactive Systems Conference
 Ishan Chatterjee, Olga Khvan, Tadeusz Pforte, [Richard Li](#), Shwetak Patel
- 2020 **A Real-Time Eating Detection System for Capturing Eating Moments and Triggering Ecological Momentary Assessments to Obtain Further Context: System Development and Validation Study**
JMIR mHealth and uHealth
 Mehrab Bin Morshed, Samruddhi Shreeram Kulkarni, [Richard Li](#), Koustuv Saha, Leah Galante Roper, Lama Nachman, Hong Lu, Lucia Mirabella, Sanjeev Srivastava, Munmun De Choudhury, Kaya de Barbaro, Thomas Ploetz, Gregory D Abowd
- 2020 **Optical Gaze Tracking with Spatially-Sparse Single-Pixel Detectors**
IEEE International Symposium on Mixed and Augmented Reality
[Richard Li](#), Eric Whitmire, Michael Stengel, Ben Boudaoud, Jan Kautz, David Luebke, Shwetak Patel, Kaan Akşit
- 2020 **KissGlass: Greeting Gesture Recognition using Smart Glasses**
Augmented Humans International Conference
[Richard Li](#), Juyoung Lee, Woontack Woo, Thad Starner
- 2019 **Prediction of Mood Instability with Passive Sensing**
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
 Mehrab Bin Morshed, Koustuv Saha, [Richard Li](#), Sidney K D'Mello, Munmun De Choudhury, Gregory D Abowd, Thomas Plötz
- 2019 **ScratchThat: Supporting Command-Agnostic Speech Repair in Voice-Driven Assistants**
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
 Jason Wu, Karan Ahuja, [Richard Li](#), Victor Chen, Jeffrey Bigham
- 2019 **TongueBoard: An Oral Interface for Subtle Input**
Augmented Humans International Conference
[Richard Li](#), Jason Wu, Thad Starner
- 2018 **ScratchVR: Low-Cost, Calibration-Free Sensing for Tactile Input on Mobile Virtual Reality Enclosures**
ACM International Symposium on Wearable Computers
[Richard Li](#), Victor Chen, Gabriel Reyes, Thad Starner
- 2018 **Buccal: Low-Cost Cheek Sensing for Inferring Continuous Jaw Motion in Mobile Virtual Reality**
ACM International Symposium on Wearable Computers
[Richard Li](#), Gabriel Reyes

- 2018 **WristWash: Towards Automatic Handwashing Assessment Using a Wrist-Worn Device**
ACM International Symposium on Wearable Computers
 Hong Li, Shishir Chawla, [Richard Li](#), Sumeet Jain, Gregory D Abowd, Thad Starner, Cheng Zhang, Thomas Plötz
- 2017 **EarBit: Using Wearable Sensors to Detect Eating Episodes in Unconstrained Environments**
Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies
 Abdelkareem Bedri, [Richard Li](#), Malcolm Haynes, Raj Prateek Kosaraju, Ishaan Grover, Temiloluwa Prioleau, Min Yan Beh, Mayank Goel, Thad Starner, Gregory Abowd

Lightly-Reviewed Publications

- 2025 **Towards Characterizing Knowledge Distillation of PPG Heart Rate Estimation Models**
NeurIPS Workshop on Learning from Time Series for Health
 Kanav Arora, Girish Narayanswamy, Shwetak Patel, [Richard Li](#)
- 2025 **Investigating the Use of Beacon for Cognitive Monitoring in People with Alzheimer's Disease and Lewy Body Dementia via the Critical Flicker Frequency**
AMIA Workgroup on Interactive Systems in Healthcare
[Richard Li](#), Philip Vutien, Ravi Karkar, Sabrina Omer, Michael Yacoub, Sarah Payne, Sean A Munson, James Fogarty, George Ioannou, Debby Tsuang
- 2024 **Improving Medical Outcomes through At-Home, Longitudinal Health Monitoring**
UbiComp/ISWC Doctoral Colloquium
[Richard Li](#)
- 2024 **Exploring and Characterizing Large Language Models for Embedded System Development and Debugging**
ACM CHI Late-Breaking Work
 Zachary Englhardt, [Richard Li](#), Dilini Nissanka, Zhihan Zhang, Girish Narayanswamy, Joseph Breda, Xin Liu, Shwetak Patel, Vikram Iyer
- 2024 **Opportunistic Nudges for Task Migration Between Personal Devices**
ACM CHI Late-Breaking Work
 Nikhita Joshi, [Richard Li](#), Jiannan Li, Leonardo Pavanatto, Michel Pahud, Jatin Sharma, Bongshin Lee, Hugo Romat, William Buxton, Nicolai Marquardt
- 2024 **The "Get Moving Trial": A Phase I/II RCT of Home-Based (P)rehabilitation ((P)REHAB) with ExerciseRx in Muscle-Invasive Bladder Cancer (MIBC)**
ASCO Journal of Clinical Oncology
 Hanna Hunter, Cindy Lin, [Richard Li](#), Otari Ioseliani, Leah Cantor, Elena Brewer, Samia Jannat, Karla Landis, David Bridges, Sean Munson
- 2023 **Smart-Pikachu: Extending Interactivity of Stuffed Animals with Large Language Models**
ACM UIST Student Innovation Challenge
 Toma Itagaki, [Richard Li](#)
- 2022 **Accuracy of Home Exercise Detection Using Novel Sonar Mobile Application Technology**
ACSM Medicine & Science in Sports & Exercise
 Ahmed I El-Sayed, [Richard Li](#), Parker Ruth, Libby Lavitt, Matthew Essman, Shwetak N Patel, Cindy Y Lin
- 2021 **Mobile, Hands-Free, Silent Speech Texting using SilentSpeller**
ACM CHI Late-Breaking Work
 Naoki Kimura, Tan Gemicioglu, Jonathan Womack, [Richard Li](#), Yuhui Zhao, Abdelkareem Bedri, Alex Olwal, Jun Rekimoto, Thad Starner

- 2020 **Evaluating a Novel, Portable, Self-Administered Device ("Flicker-App") that Measures Critical Flicker Frequency as a Test for Hepatic Encephalopathy in Patients with Cirrhosis**
AASLD The Liver Meeting Digital Experience
 Philip Vutien, Ravi Karkar, Richard Li, Kara Walter, Sean Munson, James Fogarty, George Ioannou
- 2016 **Tactile Teacher: Enhancing Traditional Piano Lessons with Tactile Instructions**
ACM Conference on Computer Supported Cooperative Work and Social Computing Poster
Richard Li, Yingyan Wang, Chih-Pin Hsiao, Nicholas Davis, James Hallam, Ellen Yi-Luen Do
- 2015 **Tactile Teacher: Sensing Finger Tapping in Piano Playing**
ACM Tangible, Embedded, and Embodied Interaction Demo
 Chih-Pin Hsiao, Richard Li, Xinyan Yan, Ellen Yi-Luen Do

Invited Talks

- 2025 **Designing and Deploying Next-Generation Health Technologies: Producing Actionable Insights from Novel Biomarkers**
 Host: Interactive Intelligent Systems Laboratory (University of Tokyo)
- 2025 **Designing and Deploying Next-Generation Health Technologies: Producing Actionable Insights from Novel Biomarkers**
 Host: Human-Computer Interaction Seminar (Columbia University)
- 2024 **Towards Personal Interpretation of Health Data for Improving Medical Outcomes**
 Host: VAK Sustainable Computing Lab (Northwestern University)
- 2024 **Supporting Exercise as a Medicine using Generative AI**
 Host: Computer Science & Engineering Industry Affiliates Day (UW)
- 2024 **Deploying and Examining Beacon for At-Home Patient Self-Monitoring with Critical Flicker Frequency**
 Host: Advanced Topics in Human-Computer Interaction (UW CSE 510)
- 2024 **ExerciseRx-CP: At-Home Sensor-Driven Strengthening**
 Host: Cerebral Palsy Grand Rounds (Seattle Children's Hospital)
- 2023 **Deploying and Examining Beacon for At-Home Patient Self-Monitoring with Critical Flicker Frequency**
 Host: Computer Science & Engineering Industry Affiliates Day (UW)
- 2022 **Sensing Device Formations using Ultra-Wideband Radios**
 Host: Human-Computer Interaction Seminar (Microsoft Research)
- 2022 **Critical Flicker Frequency for Minimal Hepatic Encephalopathy**
 Host: Hepatology Grand Rounds (Veterans Affairs Puget Sound Medical Center)

Teaching

- Fall 2024 **Embedded Systems Capstone (UW CSE/EE475, EE 542) - Instructor**
 Prepared and taught lectures on advanced embedded systems topics (i.e., PCB design, wireless communication, on-device machine learning). Guided 9 team projects from ideation to presentation. 36 students, overall rating of 4.5/5.
- Fall 2021 - Spring 2025 **Ubiquitous Computing Seminar (UW CSE 599U) - Instructor**
 Organized weekly seminar presentations on literature from ubiquitous computing, human-computer interaction, and mobile systems.
- Winter 2020 **Introduction to Human-Computer Interaction (UW CSE 440) - Teaching Assistant**
 Conducted lab sessions and graded assignments. 32 students.

Spring 2015 - Spring 2016	Technical Interview Prep (GT TIP) - Founder and Instructor Designed curriculum to prepare students for technical software engineering interviews, presented weekly lectures and organized class activities.
Spring 2015	Introduction to Computer Science (GT CS 1100) - Teaching Assistant Presented lectures on opportunities for computer science majors.

Student Mentorship

2024 - Present	Kanav Arora Deep learning for health sensing.
2022 - Present	Jason Miller Communication over audio, deep learning for pose tracking.
2022 - Present	Jonathan Shu - <i>Now: Software Engineer at Amazon.com</i> <u>UW Undergraduate Research Symposium</u> . Gaze tracking, Beacon hardware.
2021 - 2024	Toma Itagaki - <i>Now: PhD Student at Columbia University</i> <u>NSF GRFP</u> . Brain-computer interfaces, health sensing.
2020 - 2021	Millicent Li - <i>Now: PhD Student at Northeastern University</i> . <u>NSF GRFP</u> , <u>UW Undergraduate Honors Thesis</u> . Brain-computer interfaces.
2020 - 2023	Annalice Ni - <i>Now: Software Engineer at Bytedance</i> <u>UW Undergraduate Honors Thesis</u> . Fabrication and sensing techniques for interactive cardboard widgets.
2020 - 2023	Eddy Zhou - <i>Now: Software Engineer at F5</i> Fabrication and sensing techniques for interactive cardboard widgets.
2020 - 2023	Victor Chen - <i>Now: MS Student at Stanford, Software Engineer at Zoom</i> Sensing and interaction techniques for low-cost VR headsets.
2018 - 2019	Sophia Sun - <i>Now: Software Engineer at Yelp, PhD student at University of Minnesota</i> <u>GT Undergraduate Honors Thesis</u> . Face sensing for low-cost VR headsets.
2018 - 2019	Ruihan Xu - <i>Now: Software Engineer at Google</i> <u>GT Undergraduate Honors Thesis</u> . Face sensing for low-cost VR headsets.

Service

Conference Organizing

ACM CHI - 2026

Paper Reviewing

Asterisk () denotes outstanding review*

ACM CHI - 2019, 2020, 2021, 2022, 2023, 2024*, 2025, 2026

ACM IMWUT - 2019, 2021*, 2022*, 2023, 2024

ACM ISWC - 2017, 2018, 2020

ACM UIST - 2021, 2022, 2025

ACM Health - 2025

NeurIPS TS4H - 2025

IEEE TVCG - 2024

ACM TEI - 2023

IEEE VR - 2022

ACM MobileHCI - 2022

IEEE ISMAR - 2020*

Student Volunteering

UbiComp - 2019

UIST - 2022

Institution Volunteering

UW Faculty Recruiting Host (HCI) - 2024

UW CSE Visit Days Area Organizer (UbiComp) - 2022 - 2024

UW CSE PhD Admissions Area Chair (HCI and UbiComp) - 2023

UW CSE PhD Admissions Reviewer - 2022 - 2024

UW CSE PhD Pre-Application Mentorship Service (PAMS) Mentor - 2021 - 2024